VisionQAI: Distilled Bottle-Cap Defect Segmentation with Threshold Analysis and Geometric Fill-Level Estimation, Deployed with TensorRT

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***Abstract*—We present and evaluate the two inspection stations of VisionQAI, a vision system for transparent beverage bottles. Station 1 inspects the closure with a YOLO11m instance-segmentation student (22.3 M parameters). It is distilled from a YOLO11x-seg teacher (62.1 M) through feature alignment at the C3k2, SPPF, and C2PSA modules, and labels six cap states (good, damaged, misplaced, missing, open, wet) on three synchronised camera views. An asymmetric quality gate turns its detections into PASS, REJECT, UNCERTAIN, or INSPECTION_FAILED. Station 2 estimates fill level from a bottle detector, a liquid-surface (meniscus) detector, and a closed-form fill ratio with configurable underfill/overfill thresholds. On the held-out test split of the public six-class cap dataset, the student matches the teacher's mAP@0.5 of 0.995 with 2.8× fewer parameters, and exceeds the teacher's mask mAP@0.5:0.95 (0.927 vs. 0.912). As a deployed FP16 TensorRT engine it reaches mask mAP@0.5:0.95 of 0.928. Replaying the production gate gives zero false accepts (0/153 defective caps) and zero false rejects (0/201 good caps). A threshold sweep shows that good-cap scores lie in $[0.856, 0.921]$, which leaves a 0.106 margin above the default pass bar of 0.75. For fill level, the meniscus detector reaches AP@0.5 of 0.923 on its test split. The median fill-ratio error caused by meniscus localisation is 0.002, and 34 of 35 verdicts agree with those computed from the annotated surface. TensorRT FP16 cuts the three-view network time from 21.0 ms (PyTorch FP32) to 7.4 ms. In a Docker stress test the complete closed loop, including synchronous audit storage of 2048-pixel frames, runs at a median of 299 ms per bottle, and the fill station at 62.8 ms.**

***Index Terms*—bottle-cap inspection, instance segmentation, YOLO11, knowledge distillation, fill-level estimation, TensorRT, threshold analysis, quality gate**

I. Introduction

Closure and fill-level defects are among the most common visual rejects on beverage bottling lines. A missing, open, or misplaced cap compromises the seal, and an underfilled or overfilled container violates declared-volume requirements. Manual inspection does not scale to line rates of several bottles per second [1]. Learned detectors are attractive, but published food-packaging inspectors often report only validation-set accuracy, without product-level error rates or deployment latency [2], [3].

This paper reports the final iteration of a capstone system and narrows its scope to what we have built and measured. It covers two stations for transparent PET bottles. Station 1 inspects caps and Station 2 measures fill level. Product types other than transparent bottles, label checks, and operator-interface features are out of scope and are not claimed. The contributions follow the project objectives:

- **Distilled cap segmentation with threshold analysis.** We distil a YOLO11x-seg teacher into a YOLO11m-seg student through L2-normalised feature alignment at three YOLO11 modules, and pair it with an asymmetric multi-view gate. We evaluate the FP32 and deployed FP16 student on the held-out split with per-class, image-level, and product-level (false-accept/false-reject) metrics, and we map how the gate's thresholds trade review load against error.
- **Dataset and artefact characterisation (O1).** We document the six-class dataset (per-split, per-class counts, capture resolution, object-size distribution) and the exported artefacts (PyTorch, ONNX, TensorRT), and we verify that the FP16 engine preserves the student's accuracy.
- **Geometric fill-level estimation (O2).** We formalise the deployed bottle/meniscus association and the trimmed fill-ratio rule, derive its sensitivity to localisation error, and measure the fill-ratio error and verdict agreement caused by meniscus localisation on held-out images.
- **Deployment optimisation (O3).** We export all three models to TensorRT FP16 engines, measure per-stage latency against PyTorch and ONNX Runtime baselines, and stress-test the complete pipeline in its Docker deployment.

II. Related Work

Food and beverage inspection. Liakos *et al.* [4] survey machine learning for food quality control and report that beverages are a small minority of the studied applications. Baokaew *et al.* [3] classify beverage fill levels with a CNN on an embedded GPU but do not localise the liquid surface.

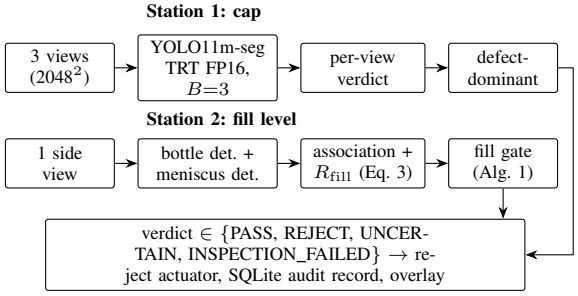


Fig. 1. The two inspection stations evaluated in this paper. Both use TensorRT FP16 engines compiled on the target GPU.

Farhangi *et al.* [5] inspect liquid bottles with rule-based edge detection and pattern matching. Liu *et al.* [2] detect five packaging defects with an attention-augmented YOLOv5. Banús *et al.* [1] compare CNN backbones for thermoformed food packages and report inline false-positive and false-negative rates on a production line, but for tray sealing rather than bottle closures or fill level. The other works report image-level accuracy without product-level error rates, and none reports end-to-end deployment latency for a multi-view bottle station.

Real-time segmentation. Ultralytics YOLO11 [6] and YOLOv8 [7] are anchor-free one-stage detectors. Their segmentation variants predict per-instance coefficients over a set of mask prototypes, following YOLACT [8].

Class imbalance and evaluation. Defect data are usually imbalanced, because compliant caps far outnumber defective ones. Common remedies are re-weighting or re-sampling the training data [9], class-balanced losses [10], and focal loss [11]. Their benefit is best judged with class-sensitive metrics: per-class precision and recall, Cohen's κ [12], and the Matthews correlation coefficient [13]. Mixup [14] and CutMix [15] regularise training, but they can blend away a defect that covers only a small part of the cap. Data quality has also been shown to matter as much as model choice in food-packaging inspection [16].

Knowledge distillation. Response distillation transfers softened logits [17]. Hint-based distillation aligns intermediate features through learned projectors [18]. Feature imitation has been applied to object detectors to compress them for deployment [19].

III. System Overview

Fig. 1 shows both stations. Station 1 receives three synchronised views of each bottle (front, rear, side). The three views pass through the network as one batch, and non-maximum suppression (NMS) is applied separately to each image. Each view is judged independently, and the bottle verdict is *defect-dominant*: a defect seen by any camera rejects the bottle. Station 2 receives one side view and runs two single-image detectors followed by a geometric rule. Both stations run the same export chain (Section VI).

TABLE I
Six-class cap dataset (instances = images; one cap per image)

Class	Train	Valid	Test
good_cap	4,242	405	201
wet_cap	828	79	39
open_cap	768	65	38
damaged_cap	648	62	31
no_cap	645	61	30
misplaced_cap	327	40	15
Total images	7,458	712	354
Source photographs	2,486	712	354

IV. Station 1: Cap Inspection

A. Dataset

We use the public `bottle_cap_sdp` v7 dataset [20] (CC BY 4.0). It contains polygon instance masks for six classes. Table I gives the counts we found on disk. The Roboflow export notes 8,530 images, but the three splits contain 8,524. Every image shows one bottle neck with one annotated cap, captured at 2048×2048 (a handful at 2048×1536) against indoor backgrounds. The training split holds 2,486 source photographs, each exported in three augmented versions (rotation $\pm5°$, brightness $\pm25\%$, Gaussian blur up to 2 px, 0.5% salt-and-pepper noise). Validation and test images are unaugmented. The masks come from a single annotator, so inter-annotator agreement cannot be measured.

The cap is not a small object in this dataset. Its median bounding box covers 9.2% of the test image (10th–90th percentile: 6.0–14.0%). After resizing to the 640×640 network input, every instance is "large" by the COCO definition (area $> 96^2$ px) [21]. The fine-scale content lies *inside* the cap: droplets on wet caps, dents on damaged caps, and the thread gap of open caps. For this reason we analyse accuracy by object size in Section VII-A but do not claim small-object detection.

B. Model and Training

The deployed model is the distilled YOLO11m-seg student [6] (Section IV-C), with 22.34 M parameters and 113.0 GFLOPs at 640 px after layer fusion. It was initialised from COCO weights and trained at 640 px with batch 32, using AdamW (initial learning rate 10^{-3}, cosine decay to 10^{-5}, 5 warm-up epochs, weight decay 5×10^{-4}). Loss gains were box 7.5, DFL 1.5, and classification 4.0. The classification gain is the mean majority-to-class count ratio (6.29), capped at 4.0. It is a single scalar and does not re-weight classes individually. Augmentation used mosaic (disabled for the last 10 epochs), copy-paste 0.5, mixup 0.15, rotation $\pm15°$, translation 0.1, scale 0.5, vertical flip 0.3, horizontal flip 0.5, HSV jitter (0.015/0.7/0.4), RandAugment, and random erasing 0.4, with mixed precision and seed 42. Training stopped early at epoch 126 (patience 30, best epoch 96) after 5.3 h on an NVIDIA RTX A5000 (Fig. 2).

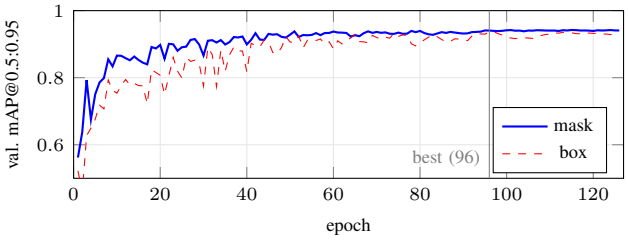


Fig. 2. Validation mAP@0.5:0.95 of the YOLO11m-seg student during distillation training.

C. Teacher–Student Feature Distillation

The teacher is YOLO11x-seg [6] (62.06 M parameters, 297.4 GFLOPs). It was trained on the same data with the student's optimiser, loss gains, and augmentation (Section IV-B) at batch 16 for the full 150 epochs (12.2 h; best epoch 139), and then frozen. Minute cap defects such as droplets, dents, and a thread gap occupy a small fraction of the cap region. Matching only the teacher's final outputs would tell the student *what* to predict but not *where* to look. We therefore distil intermediate features at three junctures of the YOLO11 architecture:

- **C3k2** (backbone output): edge and texture features, including thread and ridge geometry.
- **SPPF** (backbone–neck transition): multi-scale pooled context, so that fine cues survive spatial down-sampling.
- **C2PSA** (spatial attention): the teacher's attention map, which transfers where fine defect evidence lies.

Forward hooks capture these outputs in both networks. Learned 1×1 projectors ϕ_ℓ lift student channels to teacher width (256→512 and 512→1024), following the hint-layer approach [18]. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{YOLO}} + \alpha(t) \sum_{\ell} \left\| \hat{F}_\ell^T - \widehat{\phi_\ell(F_\ell^S)} \right\|_2^2, \qquad (1)$$

where $\hat{\cdot}$ denotes channel-wise L_2 normalisation and α ramps linearly from 0.05 to 0.5 over 20 epochs. Normalisation keeps the large, uniform background from dominating the loss, so the student matches the *pattern* of teacher activations rather than their magnitude. The ramp lets the student learn basic detection before the alignment pressure takes full effect. Teacher features are detached, so no gradient flows into the teacher. The projectors exist only during training and are discarded before export, so distillation adds no inference cost.

Outcome. Table II compares teacher and student, both evaluated in FP32 with the same Ultralytics validation on the same splits. The student matches the teacher's mAP@0.5 of 0.995 on every class with 2.8× fewer parameters, 2.6× fewer FLOPs, and a 2.8× smaller checkpoint. Its per-image inference is 7.8 ms, against 19.0 ms for the teacher. At strict IoU the two models trade places. The student's *mask* mAP@0.5:0.95 is higher than the teacher's (test 0.927 vs. 0.912; validation 0.936 vs. 0.913), and it is higher for five of the six classes; damaged caps are level (0.927 vs. 0.929). The gain is largest on the classes whose evidence is finest: wet (0.953 vs. 0.924), misplaced (0.941 vs. 0.899), and good (0.893 vs. 0.877). The teacher keeps a small lead in *box* mAP@0.5:0.95 (0.944 vs. 0.927). Station 1 decides from the cap's mask, so this is the trade the application favours. The student also has higher test recall (1.000 vs. 0.997; the teacher misses one open cap) and produces no spurious background detection, where the teacher produces one.

TABLE II
TEACHER VS. DISTILLED STUDENT (FP32, 640 PX)

	Teacher YOLO11x-seg	Student YOLO11m-seg
Parameters (M)	62.06	22.34
GFLOPs @640	297.4	113.0
Checkpoint size (MB)	124.8	45.2
Inference per image (ms)*	19.0	7.8
Test split (n=354)		
Box P / R	0.995 / 0.997	0.998 / 1.000
Box / mask mAP@0.5	0.995 / 0.995	0.995 / 0.995
Box mAP@0.5:0.95	**0.944**	0.927
Mask mAP@0.5:0.95	0.912	**0.927**
Per-class mask mAP@0.5:0.95 (test)		
damaged / good	0.929 / 0.877	0.927 / **0.893**
misplaced / no_cap	0.899 / 0.913	**0.941** / 0.915
open / wet	0.931 / 0.924	0.935 / **0.953**
Validation split (n=712)		
Box / mask mAP@0.5:0.95	0.946 / 0.913	0.938 / 0.936

*PyTorch FP32 batch 1, Ultralytics val; teacher on its training GPU.

TABLE III
PRELIMINARY WHOLE-IMAGE CAP CLASSIFIERS (SEPARATE 4-CLASS DATASET, N_{test}=308). GPU: PYTORCH, BATCH 1; CPU: ONNX RUNTIME INT8.

Model	Acc. %	mF1 %	Dmg F1	Par. M	GPU ms	CPU ms
MobileNetV3-L [22]	96.43	94.75	0.857	4.21	9.36	4.77
EfficientNetV2-S [23]	95.78	93.96	0.831	20.18	38.86	20.82
ShuffleNetV2 [24]	94.81	92.84	0.794	1.26	8.55	3.48
ResNet-50 [25]	94.81	92.30	0.788	23.52	13.14	11.19
DeiT-Tiny [26]	94.81	92.69	0.788	5.53	12.64	7.07

D. Why a Segmentation Detector

Before choosing the architecture we benchmarked five whole-image cap classifiers on a separate four-class dataset (7,540 images; 308-image test split; image-level split). Table III summarises the results. The numbers come from a different dataset and label set, so they are not comparable with Station 1 results. They informed the design in two ways. First, even the best classifier fell below 0.86 F1 on the damaged-cap class. Second, a classifier gives no location, whereas the gate and the audit record need the cap's region (mask area, polygon, and per-view evidence). YOLO11m-seg provides class, box, and mask in one pass. Its three-view batch costs 7.4 ms on the deployment GPU (Section VI).

E. Asymmetric Multi-View Quality Gate

The network runs with a confidence floor of 0.35. For each view, the gate evaluates the rules in order: (i) no detection ⇒ INSPECTION_FAILED; (ii) any defect-class detection with score $\geq \tau_d$=0.45 ⇒ REJECT; (iii) top detection is good_cap with score $\geq \tau_p$=0.75 ⇒ PASS; (iv) top detection is good_cap with score in $[\tau_u, \tau_p)$, τ_u=0.40 ⇒ UNCERTAIN; (v) otherwise UNCERTAIN. The bottle verdict is REJECT if any view rejects, else INSPECTION_FAILED if any view failed, else UNCERTAIN if any view is uncertain, else PASS. REJECT and INSPECTION_FAILED fire the actuator. UNCERTAIN goes to human review without ejection. The thresholds are asymmetric by design: a lower bar to reject than to pass, so a doubtful cap is never auto-passed.

V. Station 2: Fill-Level Estimation

A. Models and Data

The *bottle detector* is YOLOv8s [7] (9.84 M parameters, classes bottle and cap). It was trained at 512 px for up to 150 epochs (AdamW, learning rate 10^{-3}, batch 8, mosaic 0.8, mixup 0.15) on a merged collection of four public bottle and cap sources: 2,286/417/280 images in train/valid/test, with 2,133/774/356 bottle instances. The *meniscus detector* is YOLOv8n (2.69 M parameters, single class `water_surface`). It was trained at 640 px for 110 epochs without mosaic or mixup on a merge of two public sets [27], [28]: 545/31/41 images and 1,683/54/76 surface boxes. One source set consists of frames from a video of a multi-bottle rack.

B. Association and Fill Ratio

For a frame of width W, the bottle detector runs at confidence ≥ 0.25, and the highest-scoring *bottle* box $b = (x_1^b, y_1^b, x_2^b, y_2^b)$ is kept. The meniscus detector then runs on the full frame at confidence ≥ 0.15. A candidate surface box w is associated with the bottle when its horizontal centre lies inside $[x_1^b, x_2^b]$ and either its top or its bottom edge lies inside $[y_1^b, y_2^b]$. Among associated candidates, the highest-scoring one is kept. With bottle height $h = y_2^b - y_1^b$, the bottle box is trimmed to compensate for detector padding:

$$y_1' = y_1^b + \lfloor 0.03h \rfloor, \quad y_2' = y_2^b - \lfloor 0.02h \rfloor, \quad h' = y_2' - y_1'. \tag{2}$$

The surface height is the box mid-line, clamped to the trimmed bottle, $y_w = \min\big(y_2', \max(y_1', \lfloor (y_1^w + y_2^w)/2 \rfloor)\big)$. The fill ratio is

$$R_{\text{fill}} = \frac{y_2' - y_w}{h'} \in [0, 1]. \tag{3}$$

Fig. 3 illustrates the geometry. The thresholds θ_u=0.40 and θ_o=0.85 are configurable engineering defaults, not regulatory fill limits. They come from a single configuration source (environment variables), so a plant can set its own values per product. The gate, the overlay ruler, and the health endpoint all read the same values.

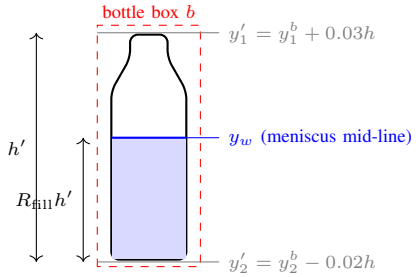


Fig. 3. Fill-ratio geometry: trimmed bottle box (gray) and meniscus mid-line (blue). Image y grows downwards.

Algorithm 1 (fill gate)
if no bottle **then** INSPECTION_FAILED (no actuation)
else if no associated meniscus **then** REJECT (empty/opaque)
else if $\min_{\theta \in \{\theta_u, \theta_o\}} |R - \theta| \leq \beta$ **and** $c_w < c_{\min}$ **then** UNCERTAIN
else if $R < \theta_u$ **then** REJECT (underfill)
else if $R > \theta_o$ **then** REJECT (overfill)
else PASS

C. Fill Gate

Algorithm 1 gives the decision rule. A borderline ratio (within $\pm\beta$=0.02 of either threshold) combined with low meniscus confidence (below $c_{\min}$=0.50) is sent to review instead of being auto-rejected. A missing bottle is treated as a conveyor gap and does not fire the actuator.

D. Sensitivity of the Rule

Differentiating Eq. (3) gives

$$\frac{\partial R}{\partial y_w} = -\frac{1}{h'}, \qquad \frac{\partial R}{\partial y_1'} = \frac{R}{h'}, \qquad \frac{\partial R}{\partial y_2'} = \frac{1 - R}{h'}. \tag{4}$$

An error of δ pixels in the meniscus therefore shifts R by δ/h'. For the median bottle in our test images (h=620 px, h'≈589 px), one pixel corresponds to 0.0017, so the $\pm$0.02 review band corresponds to about 12 px. Errors in the bottle box enter with weights R (top edge) and $1-R$ (bottom edge). Using the box mid-line instead of an edge halves the effect of a symmetric error in the meniscus box height.

VI. Deployment Optimisation

Each checkpoint is exported through ONNX to a TensorRT engine with FP16 precision [29], [30]. Reduced-precision and quantised execution can change both speed and accuracy [31], so we measure each runtime on the same hardware. The engine is compiled on the target GPU at first start and then cached under a version tag; a new tag forces a rebuild. The cap engine has a dynamic batch profile, so the three camera views share one execution, followed by separate NMS for each image. The two fill engines use a static batch of 1 at 640 px. Each engine is warmed up at load (two single-image passes, plus a three-view pass for the cap engine). Hardware is detected automatically, with fallbacks from TensorRT to ONNX Runtime [32] to PyTorch when no compatible GPU is present. The resulting files for the cap model are 45.2 MB

TABLE IV
STATION 1 SEGMENTATION ON THE HELD-OUT TEST SPLIT (n=354)

	FP32 PyTorch		FP16 TensorRT	
Class (support)	Mask $AP^{50:95}$	Box $AP^{50:95}$	Mask	Box
damaged (31)	0.927	0.952	0.931	0.958
good (201)	0.893	0.904	0.894	0.903
misplaced (15)	0.941	0.930	0.942	0.930
no_cap (30)	0.915	0.922	0.915	0.922
open (38)	0.935	0.928	0.933	0.929
wet (39)	0.953	0.929	0.953	0.930
All: $mAP^{50:95}$	0.927	0.927	0.928	0.929
All: P / R	0.998 / 1.000		0.998 / 1.000	
Valid: $mAP^{50:95}$	0.936	0.938	0.936	0.937

(PyTorch checkpoint), 90.5 MB (ONNX, FP32), and 49.1 MB (TensorRT FP16 engine).

All latency measurements were taken inside the production Docker image on one workstation: NVIDIA RTX A5000 (24 GB), Intel Xeon W-2295 (36 threads), CUDA 12.1, PyTorch 2.4.1, TensorRT 10.16.1, Ultralytics 8.4.192, ONNX Runtime 1.31.1 (CPU provider only). Each GPU configuration was warmed up with 5 calls and then timed over 100 calls.

VII. RESULTS

A. Cap Inspection (Held-Out Split)

Table IV lists segmentation metrics for the trained FP32 checkpoint and the deployed FP16 engine. On the test split both reach AP@0.5 of 0.995 for every class. Converting to FP16 changes overall mAP@0.5:0.95 by at most 0.0013 (at most 0.006 for any single class). The lowest strict-IoU score is for good_cap masks (0.893), where boundaries against the bottle neck are least distinct. Top-1 image classification, taking the highest-scoring detection above the 0.35 floor, is correct for 354/354 test images. On the validation split it is correct for 710/712 images (macro-F1 0.993, Cohen's κ 0.996, MCC 0.996 [13]). Both validation errors are misplaced caps predicted as open caps. These are two defect classes, so the bottle verdict is still REJECT.

Product-level error. Table V replays the production gate, image by image, with the deployed engine. Each image counts as one bottle observed by one view, because the dataset has no synchronised triplets. No defective cap is passed and no good cap is rejected. With zero observed events, the exact (Clopper–Pearson [33]) 95% upper bounds are 2.4% false-accept and 1.8% false-reject on the test split, and 1.2% and 0.9% on the validation split.

Threshold analysis. We swept $\tau_p \in [0.50, 0.95]$ and $\tau_d \in [0.25, 0.65]$ (Fig. 4). Neither false-accept nor false-reject occurs at any grid point. The defect threshold has no effect over its whole range, because the engine emits no wrong-polarity score above 0.01 on either split (no defect score on good caps and no good-cap score on defective caps). The pass threshold controls only the review load. Good-cap top scores lie in $[0.856, 0.921]$ on test (median 0.897), so review routing starts between τ_p=0.85 and 0.90. At 0.90, 34.5% of test bottles and 36.0% of validation bottles go to review. The default τ_p=0.75 leaves a margin of 0.106 below the lowest good-cap score. Defective caps score between 0.820 and 0.941 on test, 0.37 above τ_d. Accuracy is identical across bounding-box-area terciles. The only validation errors fall in the largest tercile.

Images outside the dataset. During system integration the deployed engine was also run on photographs of our own bottles, which the model had never seen in training or evaluation, and its verdicts on these were satisfactory.

TABLE V
PRODUCTION GATE REPLAYED WITH THE FP16 ENGINE (ONE VIEW PER BOTTLE; DEFAULTS τ_p=0.75, τ_d=0.45, τ_u=0.40)

	Valid	Test
PASS / REJECT / UNCERTAIN / FAILED	405/307/0/0	201/153/0/0
False accept (defect → PASS)	0/307 (≤1.2%)	0/153 (≤2.4%)
False reject (good → REJECT)	0/405 (≤0.9%)	0/201 (≤1.8%)
Auto-decided accuracy	1.000	1.000

Parentheses: Clopper–Pearson 95% upper bound.

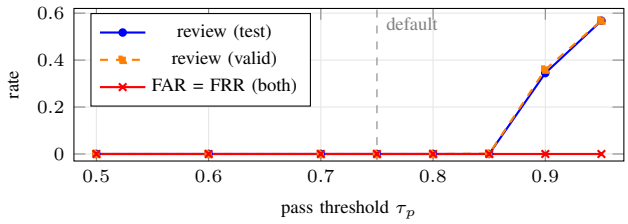


Fig. 4. Gate sweep with the FP16 engine (τ_d=0.45; other τ_d values give identical curves).

B. Fill-Level Estimation

Table VI reports both detectors on their held-out test splits at 640 px. The meniscus detector localises the surface reliably at IoU 0.5 (AP 0.923) but loosely at strict IoU (AP@0.5:0.95 of 0.458). Because Eq. (3) uses only the box mid-line, this looseness affects the ratio less than the strict-IoU score suggests. FP16 conversion leaves the meniscus detector unchanged but lowers the bottle detector's AP@0.5:0.95 from 0.578 to 0.481. The deployed bottle engine is also built at 640 px while the detector was trained at 512 px. Building the bottle engine at its 512 px training resolution is the planned remedy.

To measure how meniscus localisation affects the estimate, we ran the deployed engines on the 41 surface-test images. For each, we compared the ratio from the predicted surface with the ratio from the annotated surface, using the same predicted bottle box. Table VII gives the results. In 31 of 35 evaluable images the ratio differs by at most 0.02, and the median difference is 1 px. Verdicts agree in 34 of 35 images. The single disagreement is an overfilled bottle (annotated ratio 0.902) that was passed as normal (0.708).

Inspection of the large errors shows one failure mode. When the true surface sits high in the neck, the detector fires on a horizontal structure lower on the body, such as a moulded shoulder rib, a glare band, or the edge of a gripping hand.

TABLE VI
STATION 2 DETECTORS ON HELD-OUT TEST SPLITS (640 PX)

Model (class, n boxes)	Runtime	P	R	AP^{50}	$AP^{50:95}$
YOLOv8s (bottle, 356)	FP32	0.855	0.880	0.935	0.578
	TRT FP16	0.823	0.890	0.907	0.481
YOLOv8n (surface, 76)	FP32	0.904	0.867	0.923	0.458
	TRT FP16	0.883	0.882	0.921	0.459

TABLE VII
FILL-RATIO ERROR FROM MENISCUS LOCALISATION (41 TEST IMAGES, TRT FP16)

No bottle detected (multi-bottle rack frames)	4
Bottle found, no surface associated	2
Evaluated	35
Median $\lvert\Delta R\rvert$ / MAE / 90th pct. / max	0.002 / 0.023 / 0.022 / 0.300
$\lvert\Delta R\rvert \leq 0.02$ / ≤ 0.05	88.6% / 91.4%
Verdict agreement (under/normal/over)	34/35 (97.1%)
Gate verdicts PASS / REJECT / UNCERTAIN	27 / 7 / 1

The two missed surfaces are a tinted bottle held in a hand and a nearly full bottle. All four frames with no bottle detection come from the multi-bottle rack video, which the single-bottle design does not support.

C. Latency

Table VIII isolates the network from the rest of the pipeline. TensorRT FP16 reduces the three-view cap network time from 21.0 ms (PyTorch FP32) to 7.4 ms, a 2.8× speed-up. The single-image fill detectors drop from 8.8 and 8.3 ms to 1.3 and 1.1 ms. ONNX Runtime on the CPU needs 510 ms for the cap batch, which is too slow for this line. Measured end to end through the predictor, three 2048-pixel views take 20.1 ms. About 13 ms of that is host-side resizing and mask post-processing, which grow with batch size and are not accelerated by the engine.

Closed-loop stress test. The Station 1 stress test drives the full production path inside the container for 500 cycles: TensorRT inference, extraction of mask polygons at native resolution, the gate, a synchronous SQLite (write-ahead log) transaction that stores the three raw frames with labels, and overlay rendering. It cycles through the 354 test images, grouped three per bottle. Table IX shows the stage breakdown. The median cycle is 299 ms, which gives 163 bottles/min unpaced. In a second run paced at 150 bottles/min for 300 cycles, the line sustained 126.7 bottles/min with a 99th-percentile cycle of 448 ms. The JPEG decode of each 2048-pixel triplet falls outside the timed cycle, which is why the paced run fell short of 150. No database write failed in either run.

The dominant cost is not the network. Storing three 2048-pixel frames synchronously takes 176 ms, and the inference stage (67 ms) is mostly mask-polygon extraction at 2048^2. Station 2's service path (both engines, gate, audit, and overlay) takes 62.8 ms at the median and 111.5 ms at the 99th percentile over 500 frames.

TABLE VIII
PER-CALL LATENCY (MS, MEDIAN OF 100) INSIDE THE PRODUCTION CONTAINER

Model / input	Runtime	Network	Pre+post	Wall
Cap seg., 3 views 2048^2	PyTorch FP32	20.97	12.47	33.79
	TRT FP16	7.44	12.35	20.12
	ORT CPU FP32*	509.89	16.68	526.43
Cap seg., 3 views 640^2	TRT FP16	7.39	7.56	15.25
Cap seg., 1 view 2048^2	TRT FP16	3.90	4.38	8.59
Bottle det.	PyTorch FP32	8.77	2.11	11.21
	TRT FP16	1.26	2.07	3.60
Meniscus det.	PyTorch FP32	8.34	2.52	11.15
	TRT FP16	1.13	2.33	3.76
Fill engine (both det. + Eq. 3)	TRT FP16	p50 7.34 / p95 10.23		
Fill gate (Alg. 1)	CPU	p50 0.028 / p99 0.059		

*20 calls. Pre+post: Ultralytics preprocess + postprocess.

TABLE IX
STATION 1 CLOSED-LOOP STRESS TEST, 500 UNPACED CYCLES (MS)

Stage	p50	p95	p99
Inference (3 views, incl. mask polygons)	67.15	122.17	124.69
Quality gate	0.12	0.15	0.18
SQLite WAL audit (3 raw frames)	176.48	251.87	276.67
Overlay rendering (3 views)	41.59	43.21	43.91
Total closed-loop cycle	299.18	392.05	428.71

D. Summary Against the Stated Targets

Table X compares every measured quantity with the targets set in the earlier iteration of this project, including the ones that were missed. The false-accept and false-reject limits follow from the recall ($\geq$0.97) and precision ($\geq$0.93) targets, and the 200 ms cycle from a line rate of 300 bottles per minute.

VIII. DISCUSSION AND LIMITATIONS

Cap station. The distilled student is accurate, FP16-stable, and well separated by its gate thresholds. The threshold sweep also gives a monitoring rule for production: if the good-cap score distribution drifts down by more than about 0.1 (for example under new lighting or bottle types), the default τ_p will begin routing good bottles to review before any defect is passed. As the next benchmarking step (O1), we have identified a held-out test grouped by capture session, which will make results comparable across future model versions.

Distillation. Feature alignment at C3k2, SPPF, and C2PSA lets a 22.3 M-parameter student reproduce the 62.1 M-parameter teacher's mAP@0.5 and surpass its mask quality at strict IoU (mask mAP@0.5:0.95 +0.015 on test, +0.023 on validation), at the cost of 0.017 box mAP@0.5:0.95. This compression is what makes the 7.4 ms three-view engine possible. A comparison against a YOLO11m-seg trained without distillation would isolate the contribution of each distilled layer.

Fill level. Measured against the annotated surface, the geometric rule is accurate when the meniscus is found. Its errors are detection errors rather than errors in the arithmetic. The rule assumes one upright bottle per frame and a visible

TABLE X
ALL TARGETS AND OUTCOMES (TEST SPLIT UNLESS NOTED)

Quantity	Target	Measured	Met
Cap image-level accuracy	$\geq$0.93	1.000	yes
Cap false accept / false reject	$\leq$3% / $\leq$7%	0% / 0%	yes
Cap mask mAP@0.5:0.95 (FP16)	$\geq$0.75	0.928	yes
Cap precision / recall (FP16)	$\geq$0.93 / 0.97	0.998 / 1.000	yes
Bottle det. AP@0.5 / 0.5:0.95 (FP16)	$\geq$0.88 / 0.75	0.907 / 0.481	yes / no
Meniscus P / R (FP16)	$\geq$0.93 / 0.97	0.883 / 0.882	no / no
Meniscus AP@0.5 / 0.5:0.95 (FP16)	$\geq$0.88 / 0.75	0.921 / 0.459	yes / no
Fill verdict agreement	$\geq$0.93	0.971 (34/35)	yes
GPU inference, 3 views (network)	$<$100 ms	7.4 ms	yes
GPU inference stage, p95	$<$100 ms	122 ms	no
Closed-loop cycle, p50 (300 bottles/min)	$\leq$200 ms	299 ms	no
Fill service cycle, p50 / p99	$<$100 ms	62.8 / 111.5 ms	p50 only
CPU inference (ORT, 3 views)	$<$200 ms	510 ms	no

meniscus inside the bottle box. Our analysis isolates meniscus error only, because the surface dataset has no bottle annotations, so bottle-box error is bounded analytically (Section V-D) rather than measured. No physical fill-volume ground truth was available.

Throughput. Real-time capability depends on the I/O around the network more than on the network. Writing the audit record asynchronously and down-sampling the stored frames would remove most of the 299 ms cycle. Neither is implemented in the measured system.

Reproducibility. The datasets are public. The dataset-statistics, evaluation, and benchmark scripts and their JSON outputs accompany this paper.

IX. Conclusion

We evaluated a two-station bottle inspector built on a distilled YOLO11m-seg cap model and a geometric fill-level rule, both deployed as TensorRT FP16 engines. Feature distillation from a YOLO11x-seg teacher yields a student that matches the teacher's mAP@0.5 of 0.995 with 2.8$\times$ fewer parameters and segments caps more precisely than the teacher (mask mAP@0.5:0.95 of 0.927 vs. 0.912). On the held-out split the cap station makes no product-level errors, its thresholds have a measurable margin, and FP16 does not change its accuracy. The fill station is accurate to a median of 0.002 in fill ratio wherever the meniscus is found, and its failures are traceable to specific detection modes. TensorRT reduces network latency to 7.4 ms for three views. The closed loop is limited by synchronous image storage, not by inference.

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